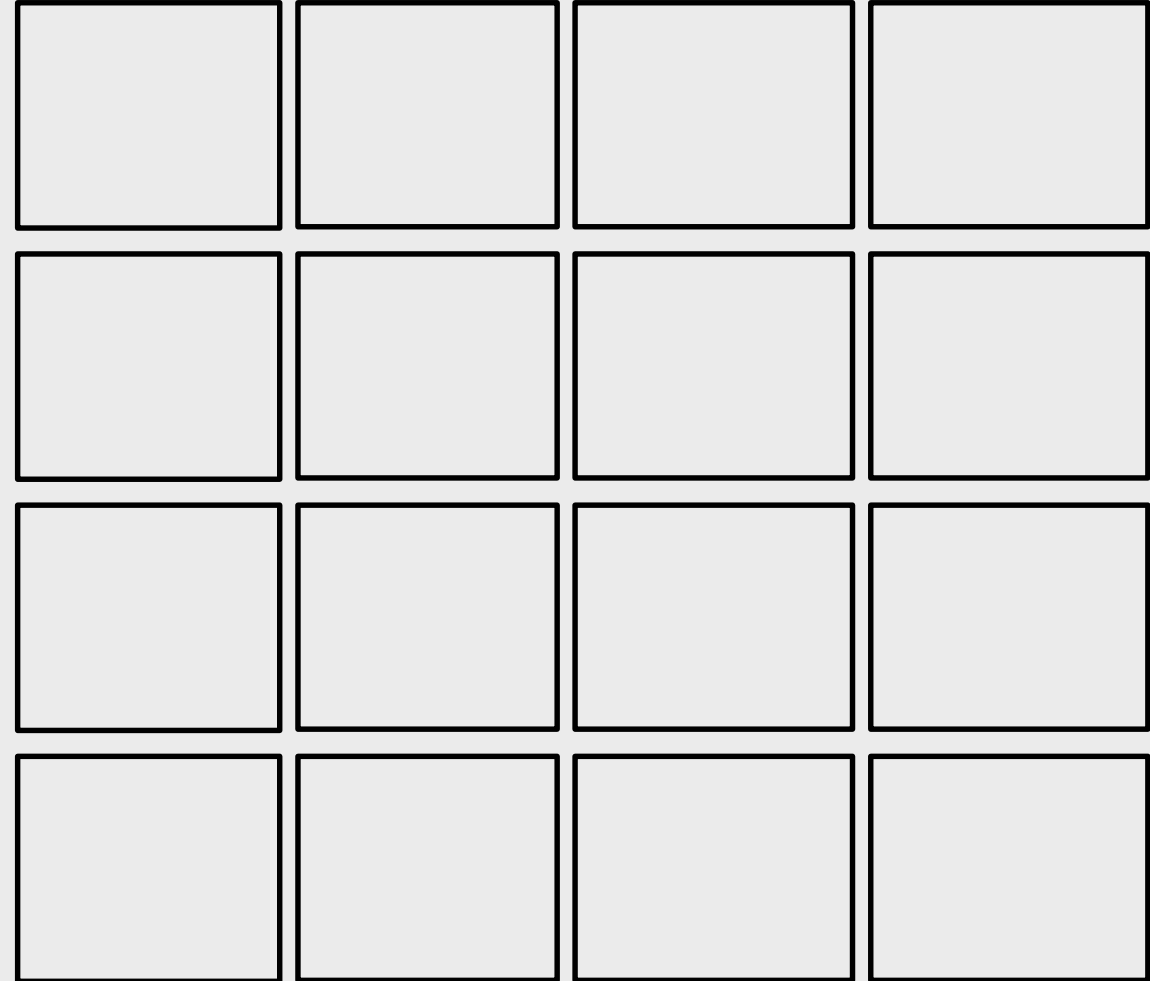




The 4-Queen Problem

Once upon a time, there was a great king in India. However, it was a matter of shame that he had 4 Queens. The Queens were so arrogant and they didn't even want to see one another. Therefore, the King built a castle of 4 x 4 rooms. However, he couldn't find a way to place the 4 Queens in 4 separate rooms, so that they couldn't see each other.

Would, you please help the King to place the Queens? Avoid placing two Queens in a same row, same column and even same diagonal rooms.





Solution of the 4-Queen Problem Using Backtracking Approach

Therefore , the king called Professor of an University to solve the 4-Queen problem. And Professor solved the 4-Queen problem in backtracking approach.

Complexity: $N\text{Queens} : O(N!)$





The 5-Queen Problem



One month later, Professor received a call from the great King to solve his 5-Queen problem. Professor, solved the 5-Queen problem in backtracking approach.



Solution of the 5-Queen Problem Using Backtracking Approach





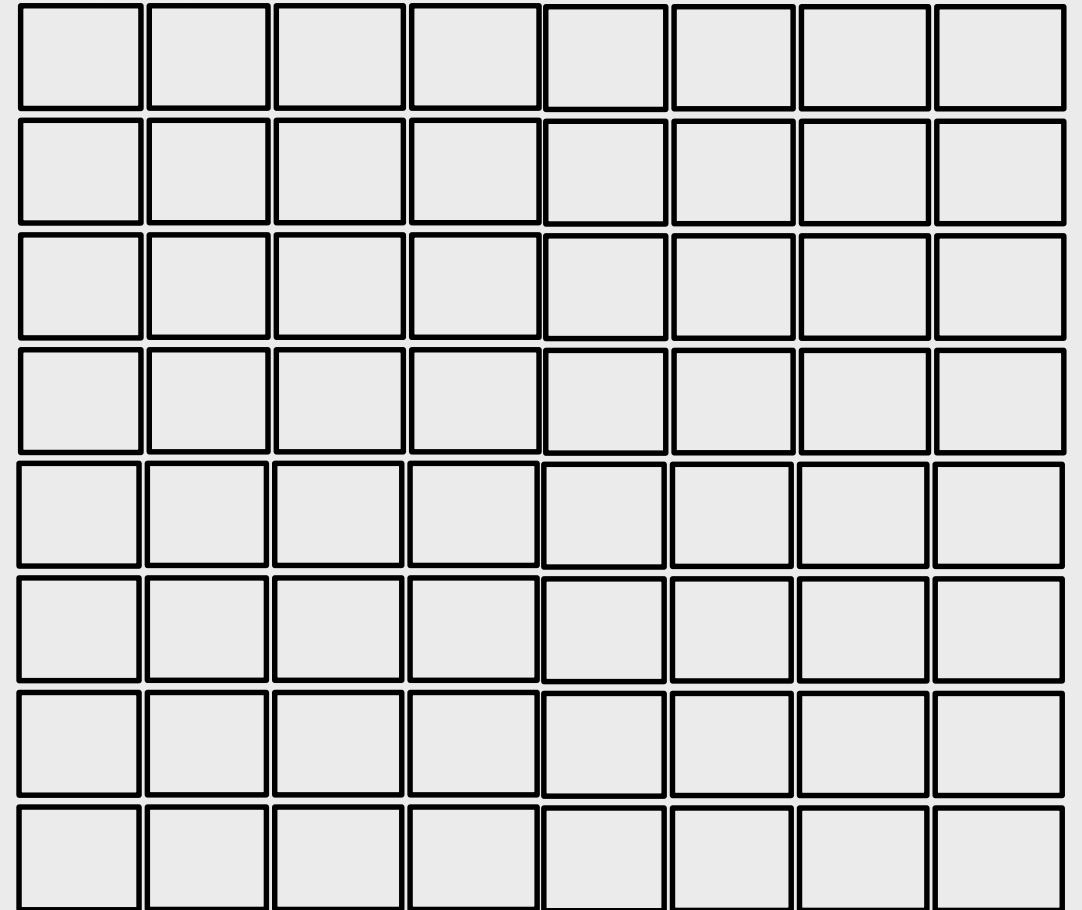
6-Queen Problem

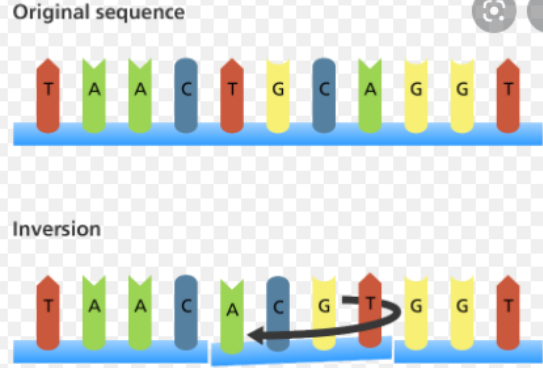


Fortunately, one month later, the King requested the professor to solve 6-Queen problem. The professor thought that the King may request him to solve 16-Queen problem within next 10 months.

Backtracking approach will not be efficient to solve the 8 or 16-Queen problems.

Therefore, professor invented Genetic Algorithm to solve the n-Queen problem.





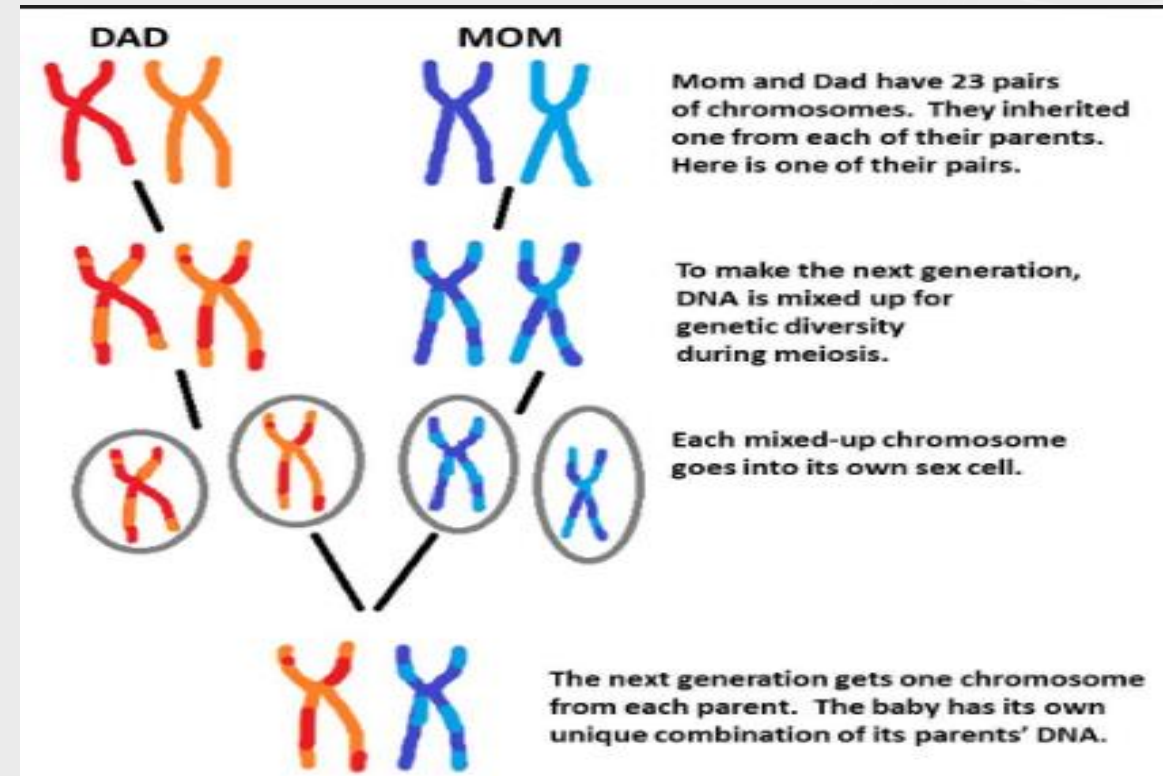
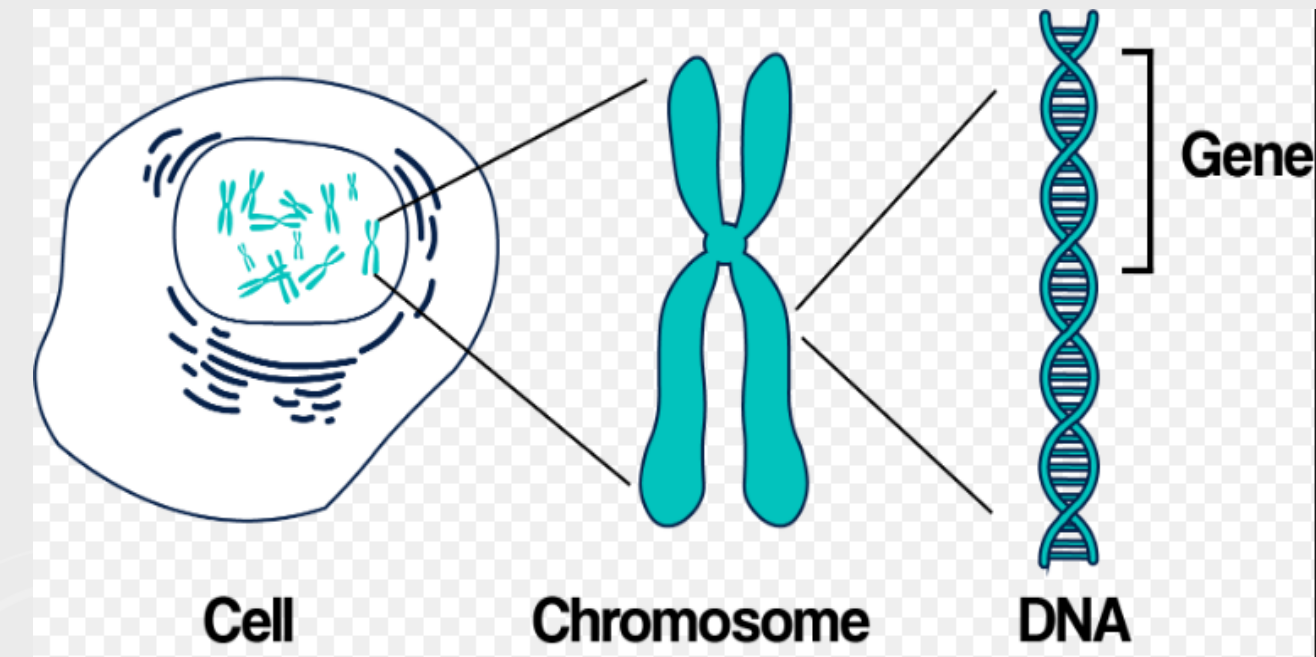
John Holland introduced **Genetic Algorithm (GA)**



Genetic Algorithm

Darwin's theory of evolution

The **genetic algorithm** is a method for solving both constrained and unconstrained optimization problems that is based on natural selection, the process that drives biological evolution.



Introduced in the 1970s by John Holland at University of Michigan

- ▶ begin with k randomly generated states (population)
- ▶ each state (individual) is a string over some alphabet (chromosome)
- ▶ fitness function (bigger number is better)
- ▶ crossover
- ▶ mutate (evolve?)

John Holland introduced Genetic Algorithm (GA)

Darwin's theory of evolution



GA: PROCEDURE

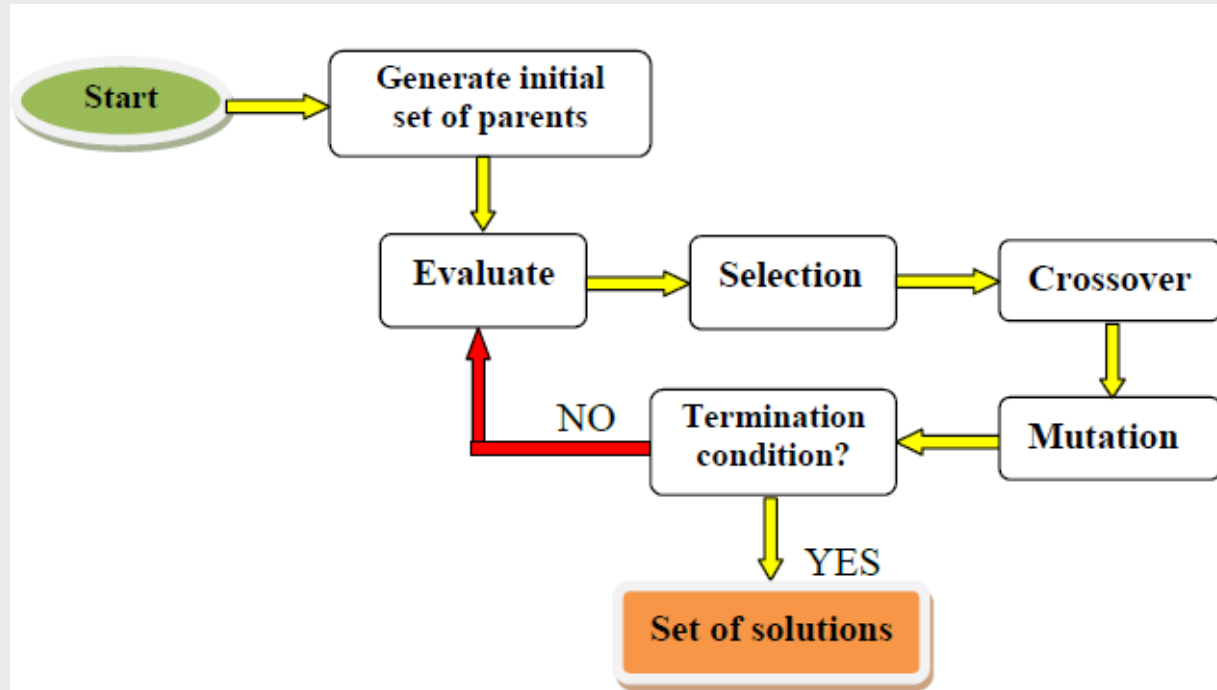


Fig.1 Process of genetic algorithm

Pseudo-code of GA:

START

 Generate the initial population

 Compute fitness

 REPEAT

 Selection

 Crossover

 Mutation

 Compute fitness

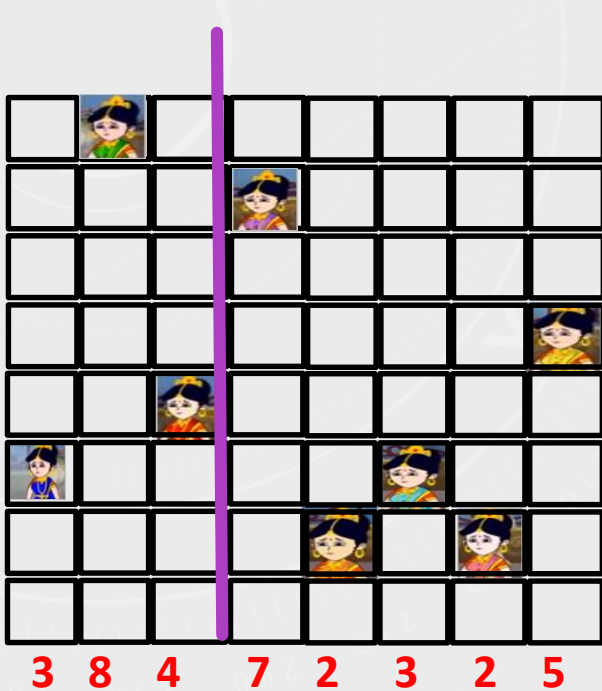
 UNTIL population has converged

STOP

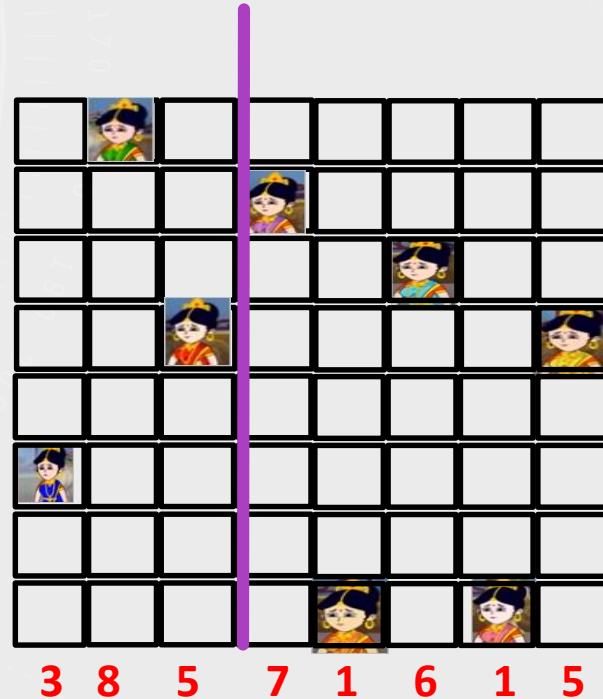


Formulation of Genetic Algorithm

John Holland introduced **Genetic Algorithm (GA)**
Darwin's theory of evolution



Fitness=28-7=21



Fitness=28-4=24

Fitness function: number of non-attacking pairs of queens

Maximum number of attacking pairs: $8 \times 7/2 = 28$

[Q1 Q2]

[Q1 Q3]

[Q1 Q4]

[Q1 Q5]

[Q1 Q6]

[Q1 Q7]

[Q1 Q8]

.....

[Q8 Q7]

Chromosome of Father: 3 8 4 7 2 3 2 5

Chromosome of Mother: 3 8 5 7 1 6 1 5

Crossover:

Chromosome from Father:

3 8 4 7 2 3 2 5

Chromosome from Mother:

3 8 5 7 1 6 1 5

Crossover point

3 8 4 7 2 3 2 5
3 8 5 7 1 6 1 5
2 4 4 1 5 1 2 4
3 2 5 4 3 2 1 3

Chromosome of Father:

3 8 4 7 2 3 2 5

Chromosome of Mother:

3 8 5 7 1 6 1 5

Offspring 1:

3 8 4 7 1 6 1 5

Offspring2:

3 8 5 7 2 3 2 5

Mutation:

Before Mutation:

Offspring 1: 3 8 4 7 1 6 1 5

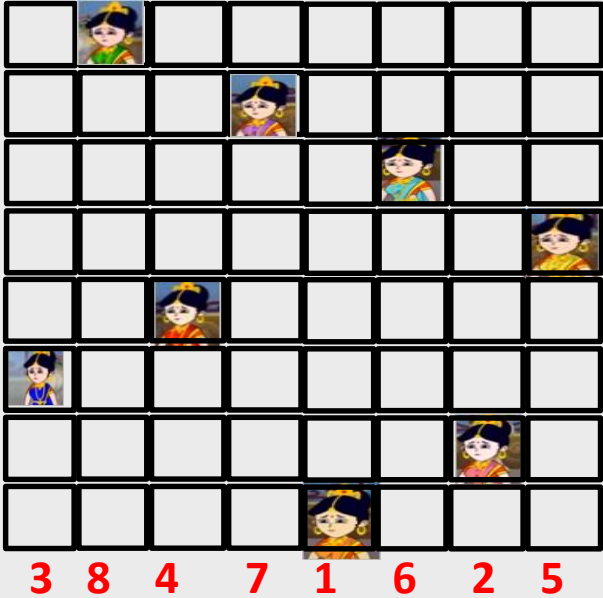
Offspring2: 3 8 5 7 2 3 2 5

After Mutation:

Offspring 1: 3 8 4 7 1 6 2 5

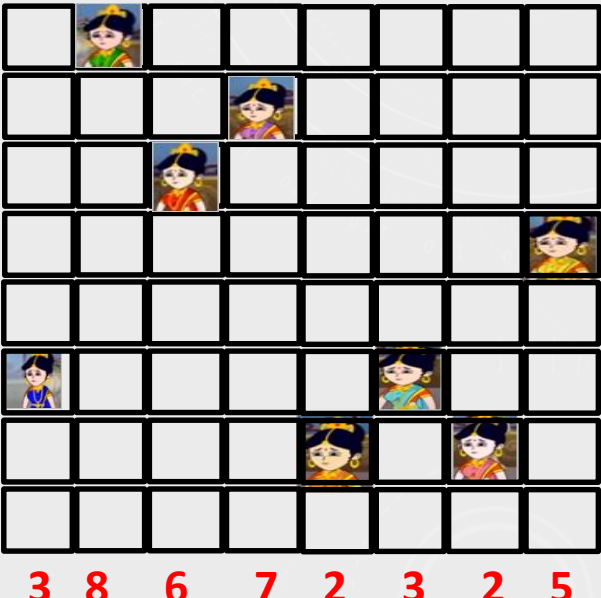
Offspring2: 3 8 6 7 2 3 2 5

Offspring 1:



Fitness=28-0=28

Offspring2:



Fitness=28-7=21

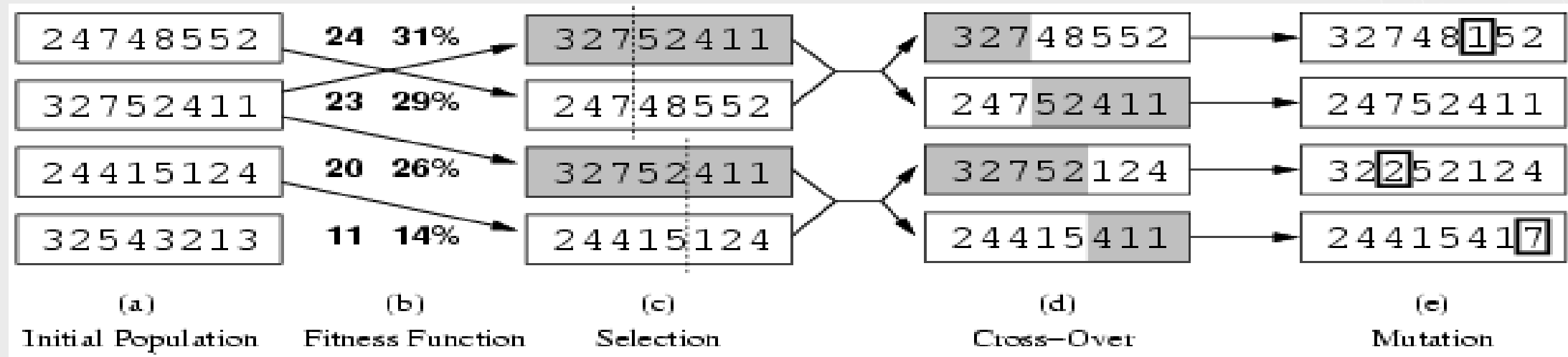
Represent states and compute fitness function.

24748552	24
32752411	23
24415124	20
32543213	11
	<u>77</u>

(a)

Initial Population

GENETIC ALGORITHMS

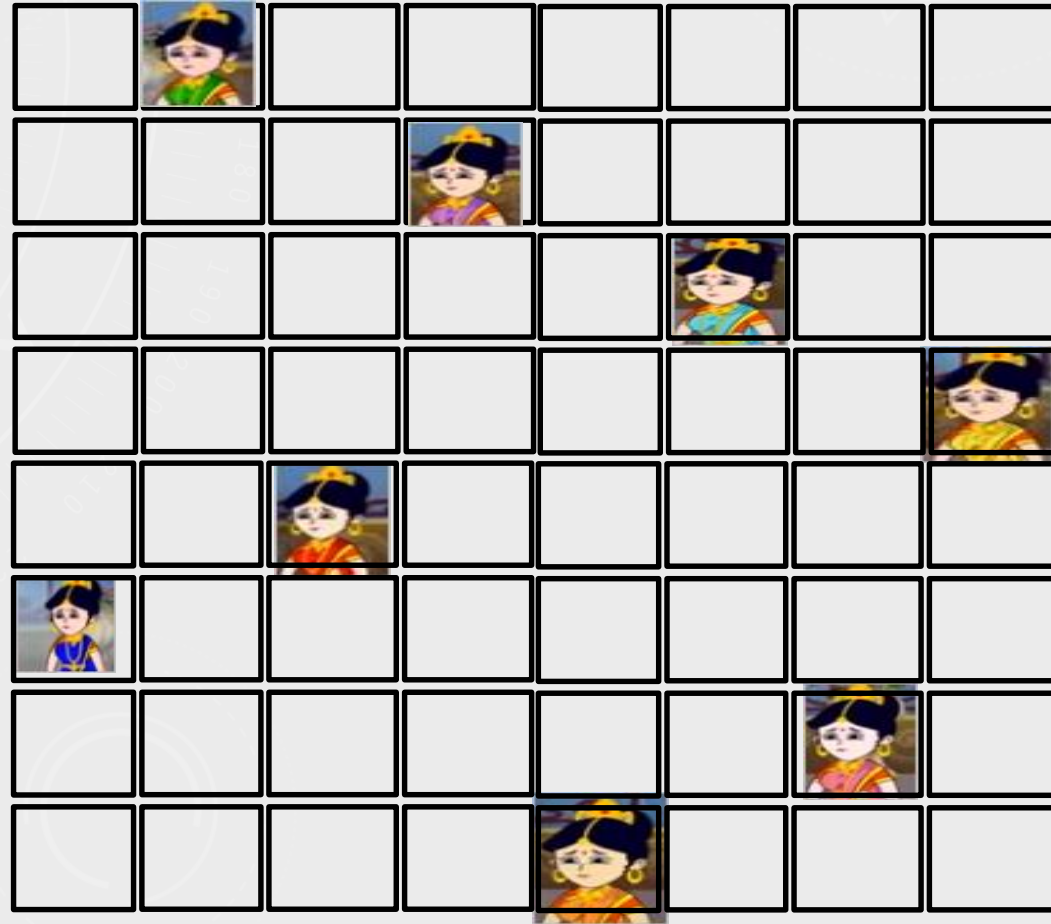


- Fitness function: number of non-attacking pairs of queens (min = 0, max = $8 \times 7/2 = 28$)
- $24/(24+23+20+11) = 31\%$
- $23/(24+23+20+11) = 29\%$ etc



Solution of 8-Queen Problem using Genetic Algorithm

John Holland introduced **Genetic Algorithm (GA)**
Darwin's theory of evolution

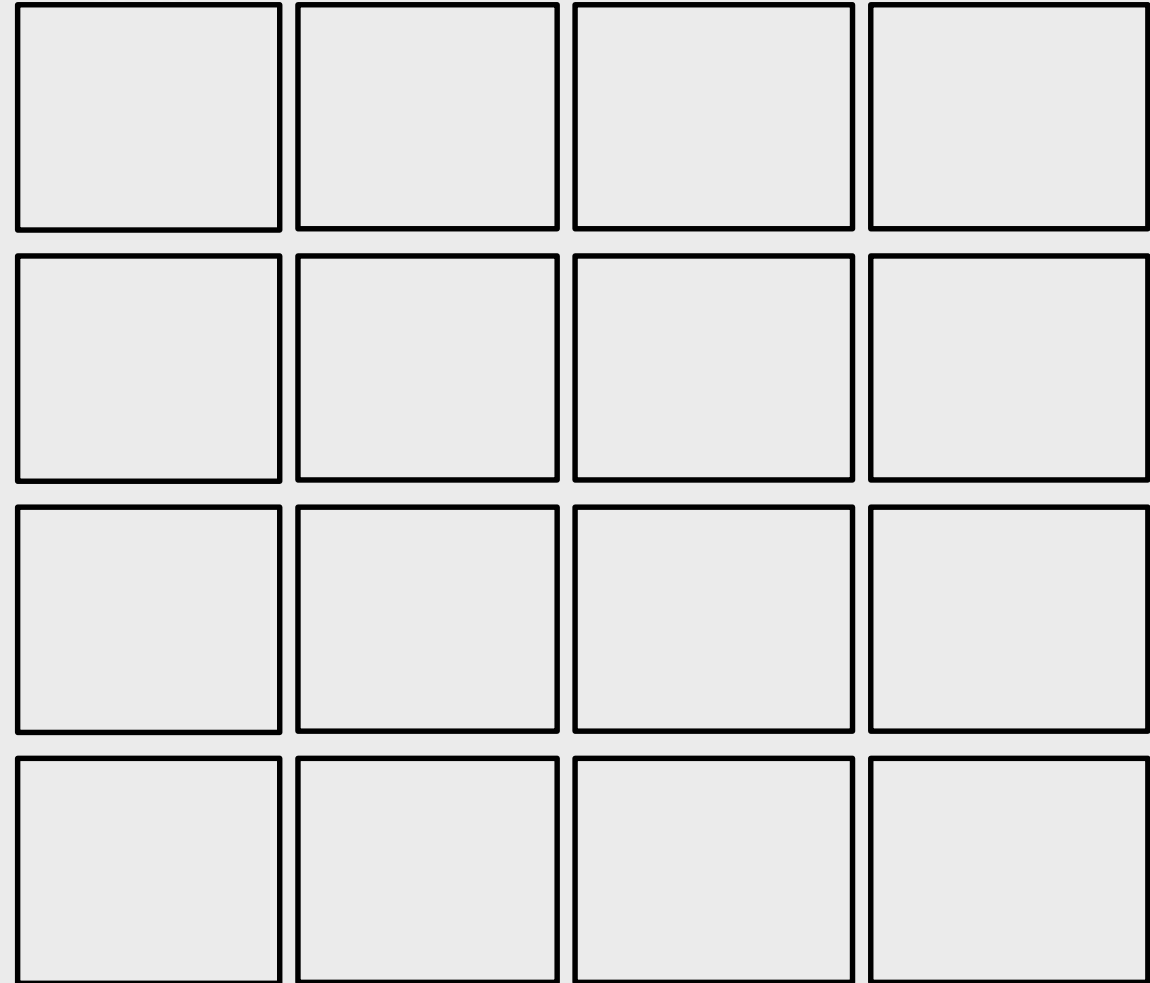




The 4-Queen Problem

Fitness function: number of non-attacking pairs of queens

What is the Maximum fitness value: ????





4-Queen Problem Using Backtracking vs GA Approach

Therefore , the king called Professor John Holland of the University of Michigan to solve the 4-Queen problem. And solved the 4-Queen problem in backtracking approach.

Complexity: NQueens : $O(N!)$

Genetic Algorithms complexity is $O(g(nm + nm + n))$ with **g the number of generations, n the population size and m the size of the individuals**. Therefore the complexity is on the order of $O(gnm)$





Try to Solve the 4-Queen Problem Using GA?



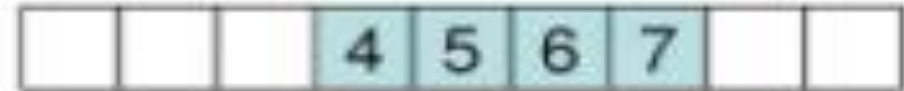
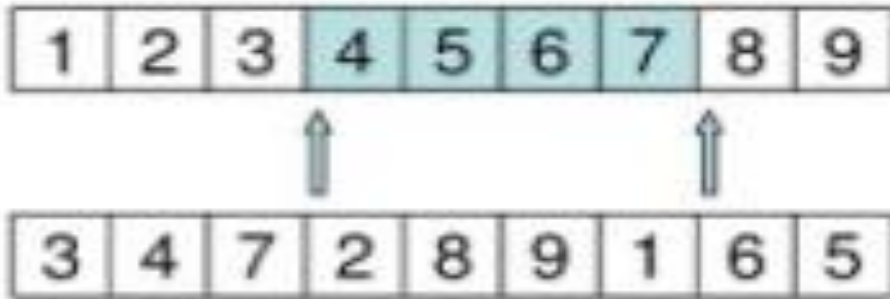
Initial Population

GA: ANALYSIS

<i>n</i>	Size of solution space (<i>n</i> !)	Number of solutions
1	1	1
2	2	0
3	6	0
4	24	2
5	120	10
6	720	4
7	5040	40
8	40320	92
9	362880	352
10	3628800	724
11	39916800	2680
12	479001600	14200
13	6227020800	73712
14	87178291200	365596
15	1307674368000	2279184
16	20922789888000	14772512
17	355687428096000	95815104
18	6402373705728000	666090624
19	121645100408832000	4968057848
20	2432902008176640000	39029188884
21	51090942171709440000	314666222712
22	1124000727777607680000	2691008701644
23	25852016738884976640000	24233937684440
24	620448401733239439360000	227514171973736
25	15511210043330985984000000	2207893435808352
26	403291461126605635584000000	22317699616364044

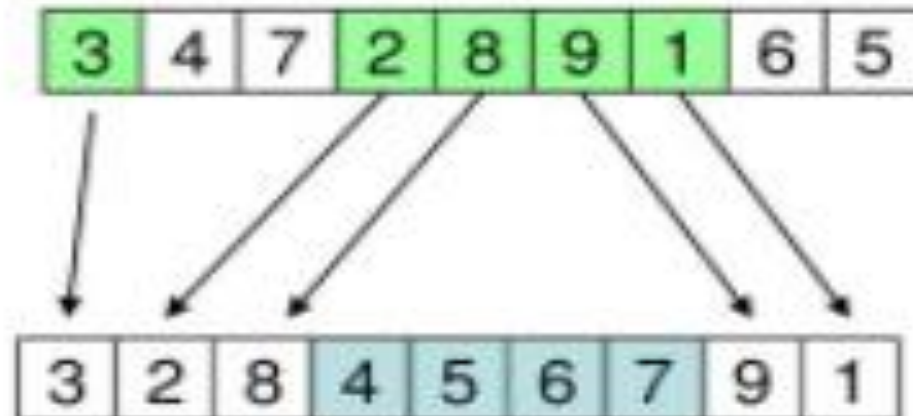
Order Crossover Example

Order crossover is similar to that of Two Point crossover, but in the case of Order Crossover, a part from first parent is copied to offspring chromosome and the remaining values are placed in the offspring by the order by which they appear in second parent.



The remaining alleles are 1 2 3 8 9.

Their order in the other parent is 3 2 8 9 1



PARTIALLY MAPPED CROSSOVER(PMX): It is

similar to that of two – point crossover. In this method, two crossover points are chosen among two parents and the genes between two crossover points are exchanged, while remaining genes are filled by partial mapping.

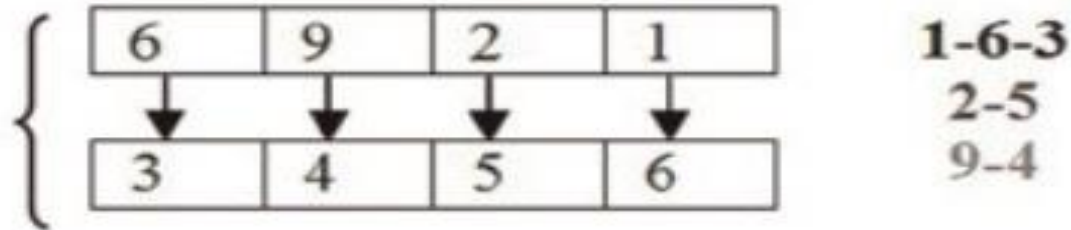
Select the sub-strings at random



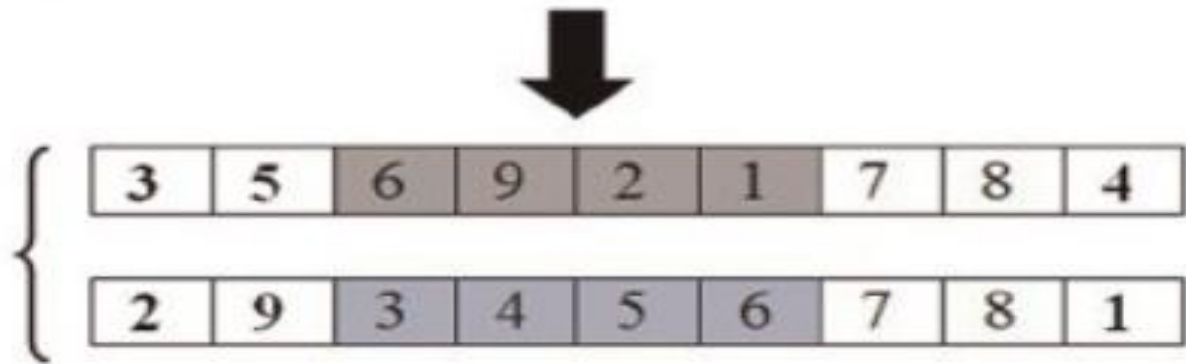
Exchange sub-strings between parents



Determine mapping relationship



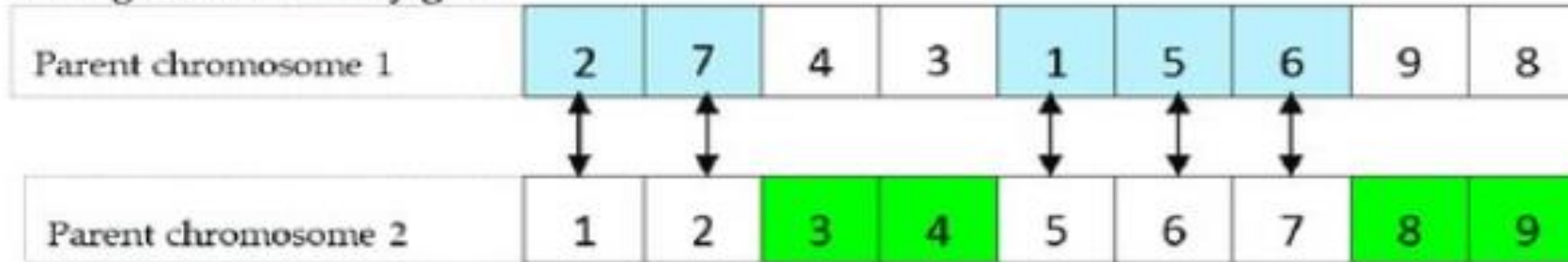
Legalize offspring with mapping relationship



CYCLE CROSSOVER (CX):

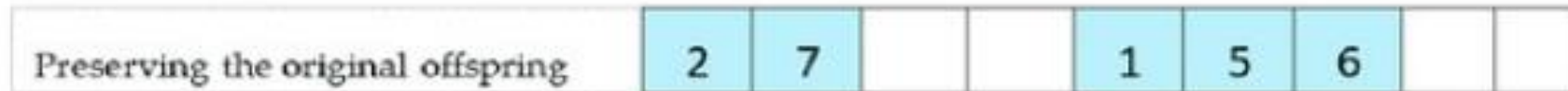
In cycle crossover, a gene from one parent will be copied into an offspring, but it should inherit the position of the other parent.

Step 1: Finding the circulatory gene



Cycle: 1—2—7—6—5—1

Step 2: The circulating gene is replicated to the offspring



Step 3: The remaining genes are identified for the offspring, and the remaining genes outside the parent chromosome cycle are used to fill with the original offspring



Step 4: Forming a descendant



SELECTION:

METHOD 1: RECURSIVE ADAM & EVE: Killing all population except the two fittest individuals.

METHOD 2: KILL HALF OF THE POPULATION: Killing the worst half of the population.

METHOD 3: THE GREAT TOURNAMENT

- A tournament essentially consists in selecting randomly a bunch of individuals and make them *fight*; and the result of this battle will decide which of them survives the purge.
- The most obvious way to decide the winner is to compare its fitness value; the victorious fighter would be he who was the greater fitness.

```
fight ← function(w1, w2, pmut, mort){  
  e1 ← meas.error(w1)  
  e2 ← meas.error(w2)  
  
  if(e1 > e2) {  
    a ← runif(1)  
    if(a < mort) vic ← w2  
    else vic ← gen.offspring (w1, w2, 1, pmut)[[1]]  
  }  
  if(e1 < e2){  
    a ← runif(1)  
    if(a < mort) vic ← w1  
    else vic ← gen.offspring (w1, w2, 1, pmut)[[1]]  
  }  
  if(e1 == e2) vic ← gen.offspring (w1, w2, 1, pmut)[[1]]  
  return(vic)  
}
```

MUTATION:

- Flip-Bit
- Swapping

Conclusion

Application areas of GA:

- Game programming
- Cloud resource allocation
- Job scheduling of operating systems
- Channel assignment in communication system
- Combinatorial optimization
- Creative design (NASA antenna)
- Operational research

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- UC Berkeley's Hans Bremermann, professor emeritus and pioneer in mathematical biology, has died at 69
- *Fogel, David B. (editor) (1998). Evolutionary Computation: The Fossil Record. New York: IEEE Press.*
- *Barricelli, Nils Aall (1963). "Numerical testing of evolution theories. Part II. Preliminary tests of performance, symbiogenesis and terrestrial life". Acta Biotheoretica (16): 99–126.*
- **How to easily find the optimal solution without exhaustive search using Genetic Algorithms**

RESEARCH

A Dynamic Scheduling Method for Collaborated Cloud with Thick Clients

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